

Introduction

My primary research question is: do time-varying acoustic features of narrative audio, including sound energy, spectral structure, and pitch, predict patterns of brain activity during naturalistic story listening? Furthermore, are these neural responses to acoustic features different during segments containing just speech, just music, or both speech and music? I plan on analyzing the “The Tunnel Under The World” from the Narratives Dataset (Nastase 2021). and extracting acoustic features using analysis libraries such as librosa. It includes functions to select sound energy, spectral features such as spectral centroid and Mel-Frequency Cepstral Coefficients (MFCCs), pitches, and spectral flux as time series that I can align to the audio timeline (with appropriate fMRI hemodynamic offset). I also plan on manually tagging the audio into three categories: speech-only, music-only, and speech + music, which can be used as grouping variables. I will then run classification, covariance, and ISC analyses as we have done in labs to evaluate the neural encoding sensitivities and identify regions of the brain where activity is best explained by the acoustic features.

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2. Setup

Below, we import all the necessary libraries for the notebook and set up paths to directories that we will pull data from.

```
Dir /tmp/final_project_results/ created.
```

2A. Load Audio

We use the `librosa` library to load the Tunnels Under the World audio. It decodes the file and returns a time-series of floats (using the default sampling rate of 22.05 Hz). This usually does not take too long.

```
Loaded /home/NEU480/datasets/narratives/stimuli/tunnel_audio.wav
Duration: 1533.77 seconds
Native SR: 22050
Time to load: 2.54 seconds
```

2B. Extract Sonic Features

We begin by extracting **sound energy root mean squared** and **spectral flux**. The former measures the effective amplitude of the sound signal, i.e. its power or loudness, by calculating the root of the mean of adjacent instantaneous values, squared. The latter detects how quickly a signal's power spectrum changes over time, thereby capturing when sounds are onset and with what strength.

	time	rms_energy	spectral_flux
0	0.00000	0.001695	0.000000
1	0.02322	0.001705	0.000000
2	0.04644	0.001722	0.000000
3	0.06966	0.001150	0.121475
4	0.09288	0.021806	0.017866
5	0.11610	0.054559	0.130707
6	0.13932	0.067246	4.359818
7	0.16254	0.087558	9.529262
8	0.18576	0.114422	4.745287
9	0.20898	0.137655	1.467012

Some other features we are looking at include **spectral centroids** and **Mel-Frequency Cepstral Coefficients (MFCCs)**, which inform our understanding of the audio's spectral structure. The centroids, per their name, indicate where most of the sound's energy is concentrated within its spectrum and measures the perceived "brightness" or "timbre" of the sound. MFCCs attempt to mimic how ears perceive sound (and therefore how the cortical region integrates diverse sonic structures) across 13 variables, where the first captures sonic energy, 2-4 the broad shape of the spectrum, and 5-13 various finer details from harmonies to consonants.

Lastly, we also extract **pitch**, important for both speech and music. We take the fundamental frequency from the audio, which will be used later for aligning with neural data from regions that may selectively respond to pitch.

Note: This steps tends to take ~10 minutes to complete.

Time to load: 887.50 seconds

Now, we combine and time-align everything by converting from real time to TRs, using the dataset-proposed TR of 1.5. For the potentially multiple frames that fall within a single TR, we take their mean.

Before we begin neural alignment, we want to redefine the sound energy column. The human ear perceives loudness logarithmically, so we will convert the linear RMS energy into a decibel (dB) scale to better represent auditory perception. To avoid $\log(0)$ issues when there's pure silence, we add a miniscule constant.

2C. fMRI Preprocessing

While brain data preprocessing steps have already been taken, such as motion correction, scanner noise, etc., we still have to convolve our audio data to the brain. Specifically, whereas `librosa` notes a potential peak in sound energy instantaneously, the BOLD fMRI signal peaks ~5-6 seconds after that same peak. Therefore, we need to convolve the sonic features with a hemodynamic response function that simulates the brain's delay, which allows us to properly compare the two.

We standardize our values using Z-scoring. Note that features like F0 (Hz) and Sound Energy (dB) have different scales, and later regression may unfairly weight features simply because of larger raw numerical values.

The last thing we want to do for alignment is to note the 23 seconds of silence after the stimulus where the scanner kept running. Since that is present in the brain data, making up a total of 1040 TRs, we need to pad this dataframe with 17 TRs, since the audio was 1534 seconds long in comparison to the 1560 seconds of scanner runtime.

2D. Tagging

As our last step of preprocessing, we want to enable experimental analysis on our "speech" vs. "music" hypothesis. This means we have to tag the audio with TR-aligned labels that correctly represent speech, music, or both.

A hybrid approach will be used:

1. inaSpeechSegmenter - a convolutional neural network-based library that splits audio into speech, music, and noise. It outputs a table in the format (label, start, end), where the latter two fields are in seconds. However, this does not natively output a 'both speech & music' label.
2. Gentle - a time-aligned transcript with word-level start and end labels, also in seconds. This can capture speech labeling.

So, a 'both speech & music' label can be created by overlapping the two outputs.

Note: Segmenter() fails to run on Adroit because it does not come installed with its neural network weights to save space and attempts to download it from a URL. Since Adroit is offline, open a login node that has internet access and paste the following line to cache the weights in your home directory:

```
python -c "from inaSpeechSegmenter import Segmenter; Segmenter()"
```

This also takes quite a while (~20 min) to run.

```
E0000 00:00:1778692617.795387 36040 cuda_platform.cc:52] failed call to cuInit: INTERNAL: CUDA error: Failed call to cuInit: UNKNOWN ERROR (303)
```

```
1997/1997 - 42s - 21ms/step
1596/1596 - 168s - 105ms/step
Time to load: 2401.99 seconds
  label  start   end
0   male   0.00   2.42
1 noEnergy  2.42   3.20
2   male   3.20   4.80
3 noEnergy  4.80   5.38
4   music   5.38  13.02
  label  start   end
4   music   5.38  13.02
29  music  66.82  78.54
43  music 121.92 124.94
45  music 134.10 135.58
86  music 209.00 215.34
```

Gentle transcripts have a specific JSON format that we will parse now. Additionally, as the dataset metadata mentioned 126 unsuccessfully localized words, this will also show in the transcript, specifically in the `case` field as `not-found-in-audio`. Looping through every word looking for the `start` and `end` fields will throw a key error for those words, so we begin by filtering them out.

Total words: 3435

Successfully aligned speech blocks: 3309

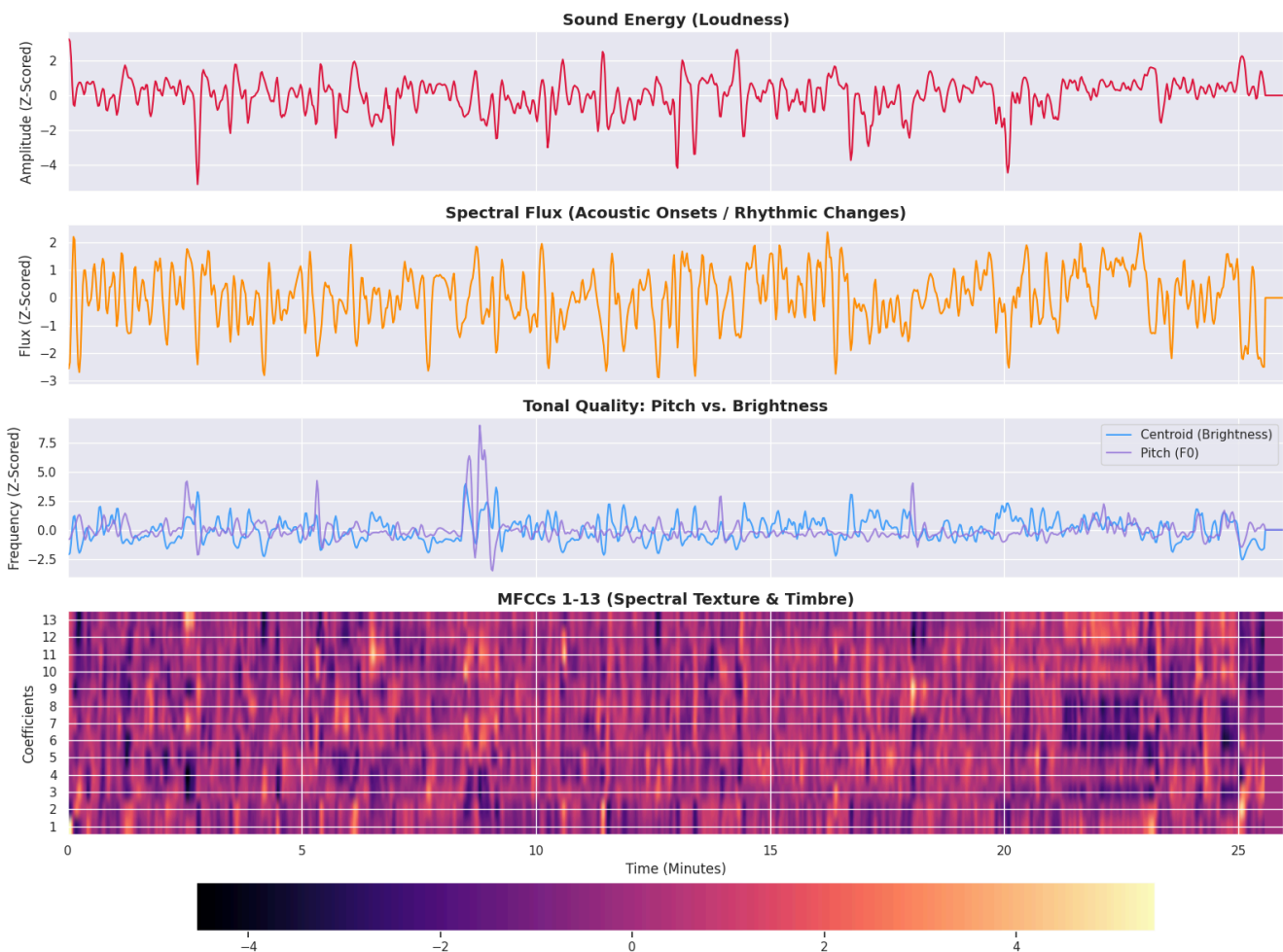
Now we can align the inaSpeechSegmenter data with the Gentle transcript.

Tagging complete. Final matrix shape: (1040, 24)

	tr_index	category	is_speech	is_music	is_speech_music
0	0	speech	True	False	False
1	1	speech	True	False	False
2	2	speech	True	False	False
3	3	music	False	True	False
4	4	music	False	True	False

3. Initial Testing

We can finally begin our running analyses. We start with some basic visualizations to see the sonic structure of the Tunnel audio. Note that MFCCs are dimensionless numerical values.



3A. Encoding

Let's try aligning these features to the brain. We will focus primarily on the auditory cortex in the superior temporal lobe (Heschl's Gyrus) and the superior temporal gyrus. To get the BOLD signals for these specific ROIs, we align the brain to the Harvard-Oxford cortical

atlas and extract those patches. For now, let's work with just one subject to confirm some preliminary results to assure us of how we parsed the audio data.

`[fetch_atlas_harvard_oxford]` Dataset found in `/home/NEU480/datasets/nilearn_c`

Heschl's Gyrus index: 45

STG Anterior index: 9

STF Posterior index: 10

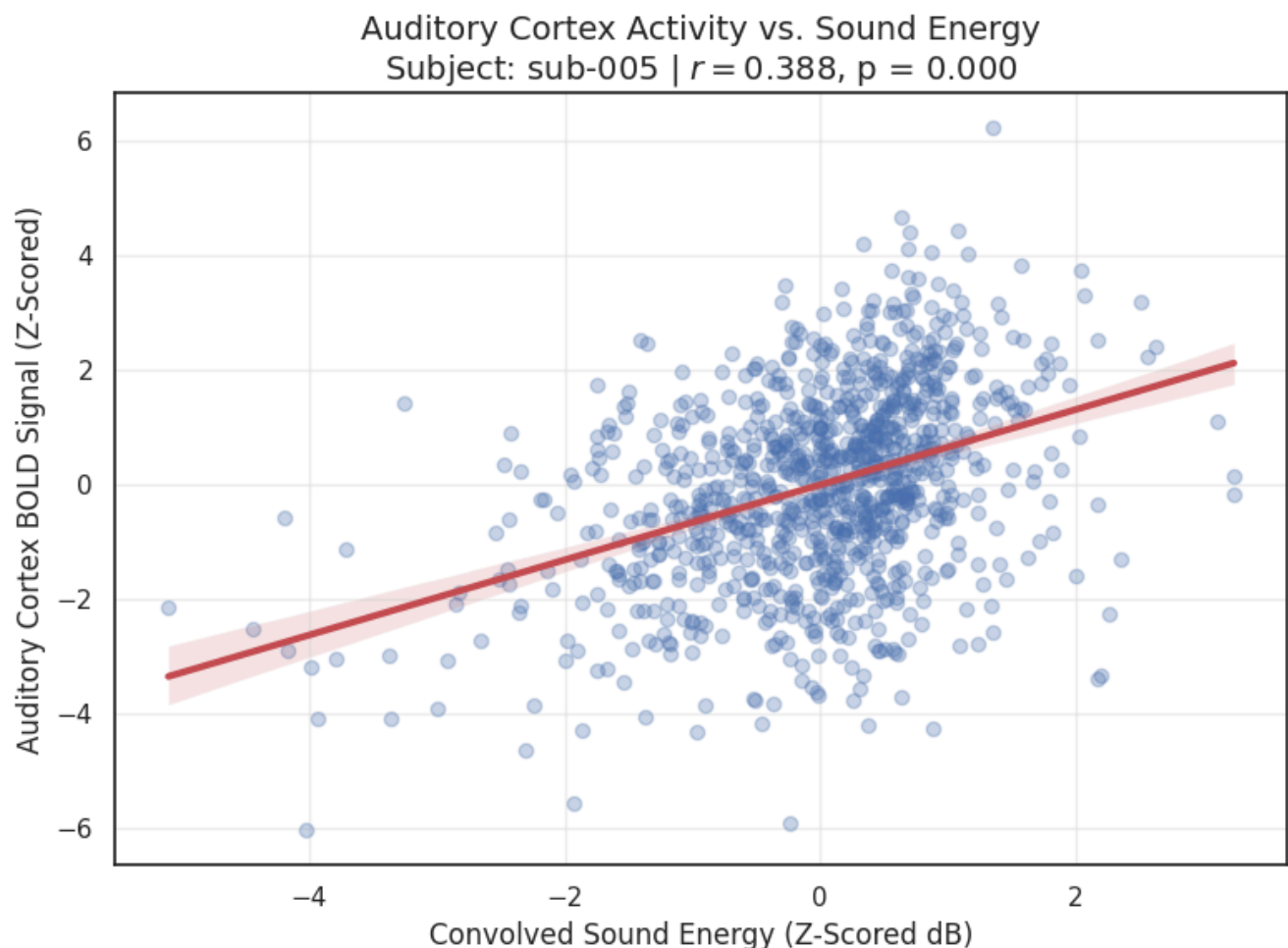
Processing subject: `/home/NEU480/datasets/narratives/derivatives/afni-nosmooth/sub-005/func/sub-005_task-tunnel_space-MNI152NLin2009cAsym_res-native_desc-clean_bold.nii.gz`

Extracted time series shape: (1040, 48) (Expected: 1040 TRs, 48 Regions)

Our first experiment is to correlate sound energy with BOLD activity. Schmalzle et. al. (2022) found that in naturalistic audio listening, temporal changes in sound energy predicted moments of heightened attention and synchronicity. While we explore synchronicity later using ISC, we test here if there is indeed a positive correlation between energy and brain activity.

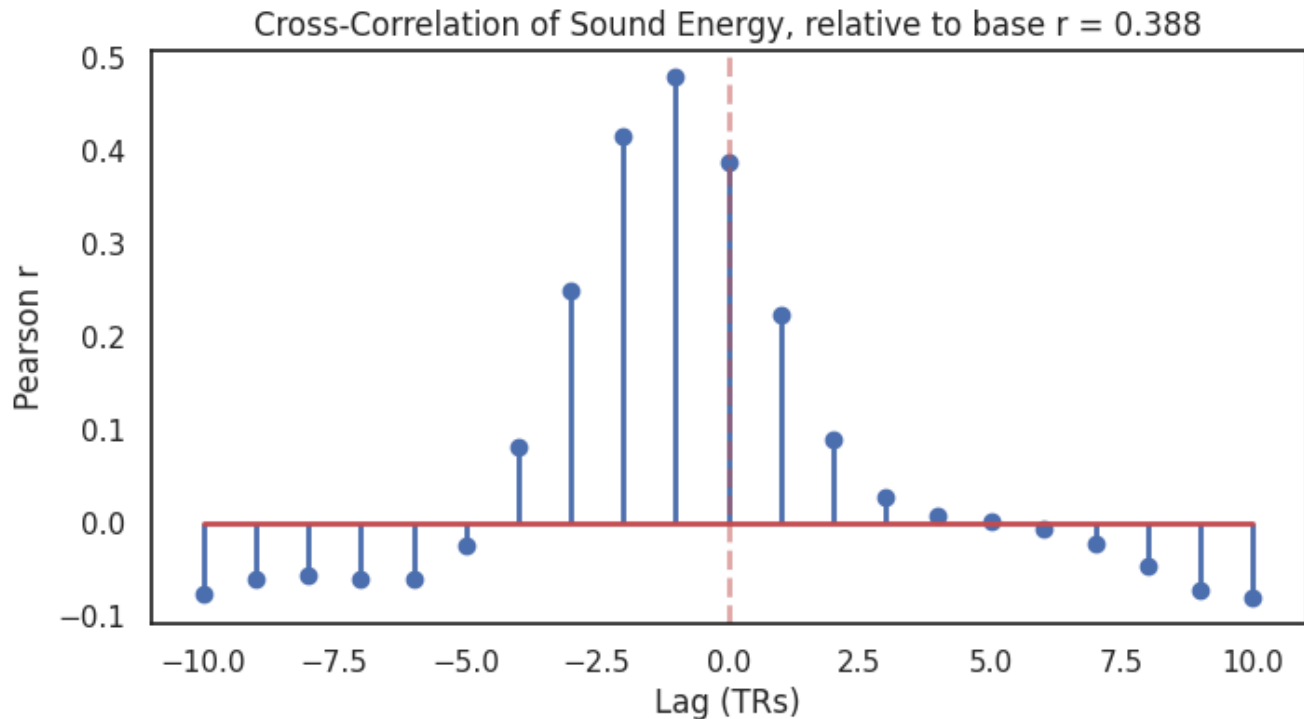
Correlation (r): 0.3881

P-value (p): 9.9751e-39



This is a good result! With a r -value of 0.3881 and a significant p -value, we have shown that there is a strong relationship between sound energy and a participant's brain activity in the auditory cortex. However, in another preliminary test I ran (not included in this notebook), I had to measure cross-correlation, i.e. the correlation between two signals if one was displaced relative to the other, and found the following results:

Base Correlation: $r=0.3881$, $p=9.9751e-39$



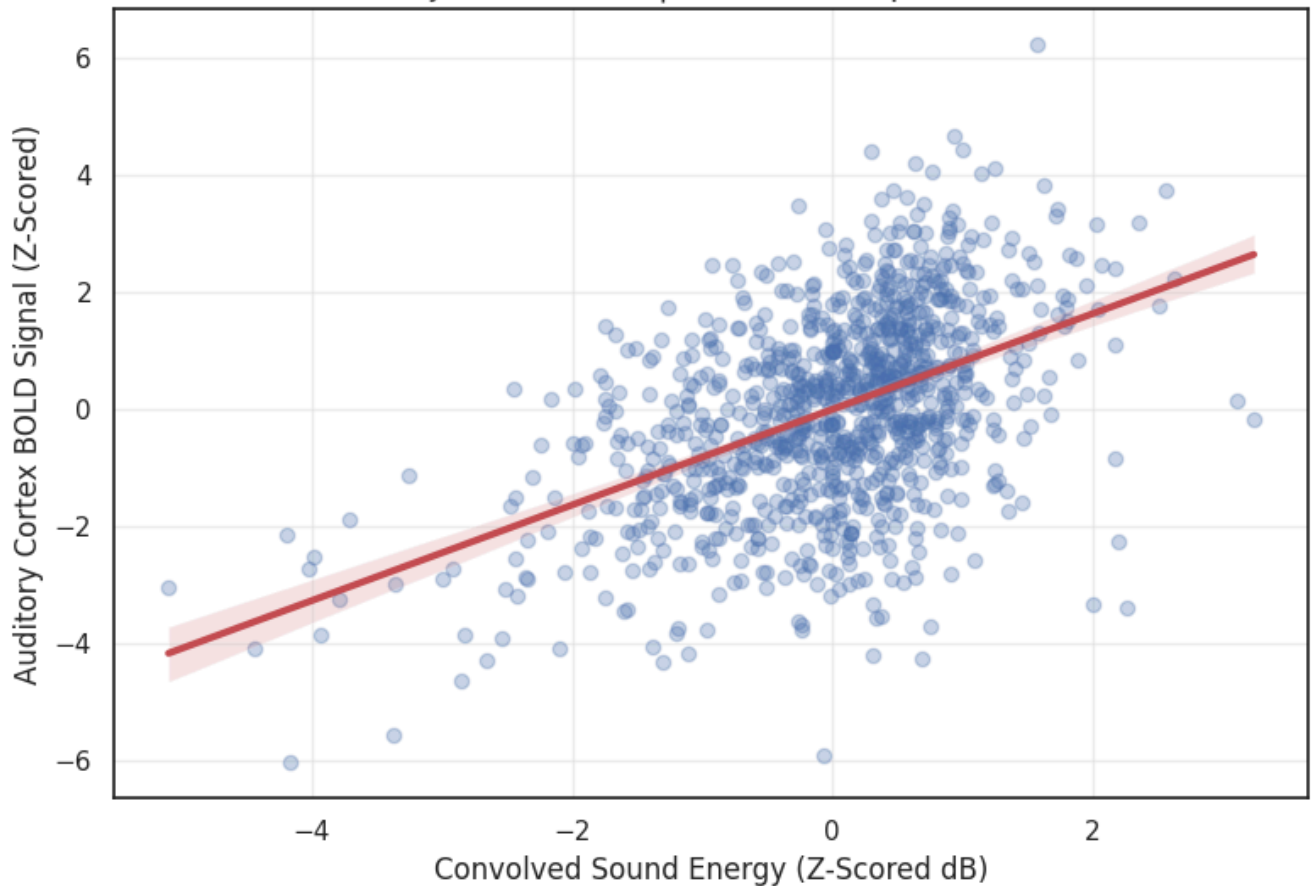
While this does follow the general HRF basis function that we might expect to see, with decreasing correlation in either direction of the peak, it seems that the time alignment preset by the dataset may have been too much. In investigating this phenomenon, I came across Handwerker et. al. (2004), which found that hemodynamic responses varied across subjects and crucially, brain ROIs. Time-to-peak or time-to-onset had an observed 2.5 seconds window of misestimates. The general HRF convolution assumes an averaged response time for the entire brain, but with this differential HRF, it could be the reason why the auditory cortex's correlation seems to peak an entire TR earlier. To capture this important characteristic, we need to temporally shift everything 1 TR back.

Applied -1 TR physiological shift. Final shape: (1040, 24)

With the correct temporal padding, let's try the previous sound energy vs. activity in Heschl's Gyrus again.

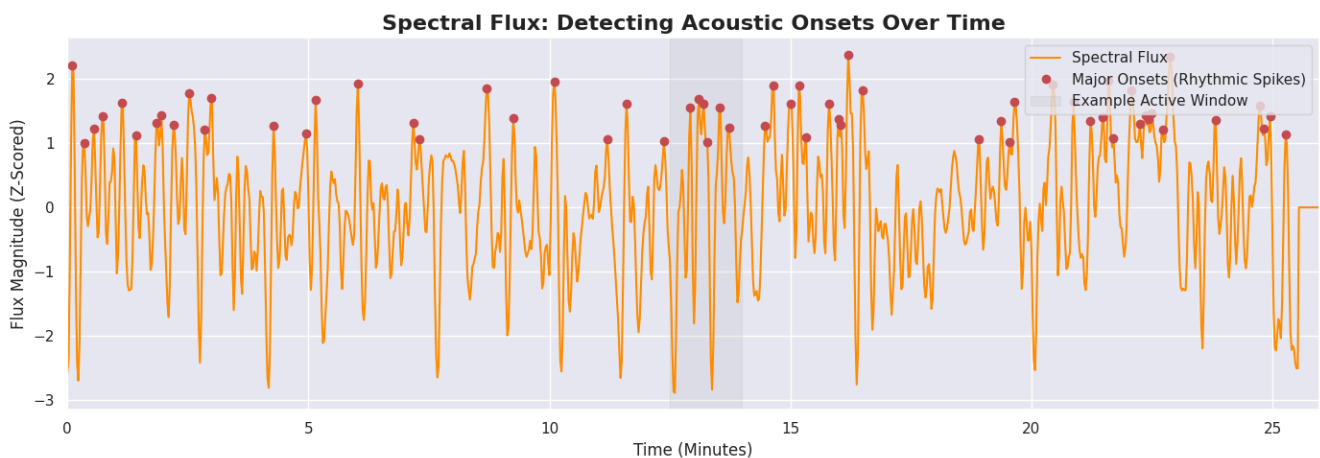
Corrected Correlation: $r=0.4807$, $p=3.0273e-61$

Heschl's Gyrus Activity vs. Sound Energy
Subject: sub-005 | $r = 0.48071$, $p = 0.000$



Much better! Our correlation went from 0.3188 to 0.4807, a huge increase. We will use this correctly aligned data for the rest of the analyses, but this also shows why it is important to verify previous results. Though we did have a positive correlation, we expected a slightly stronger relationship, which we now have.

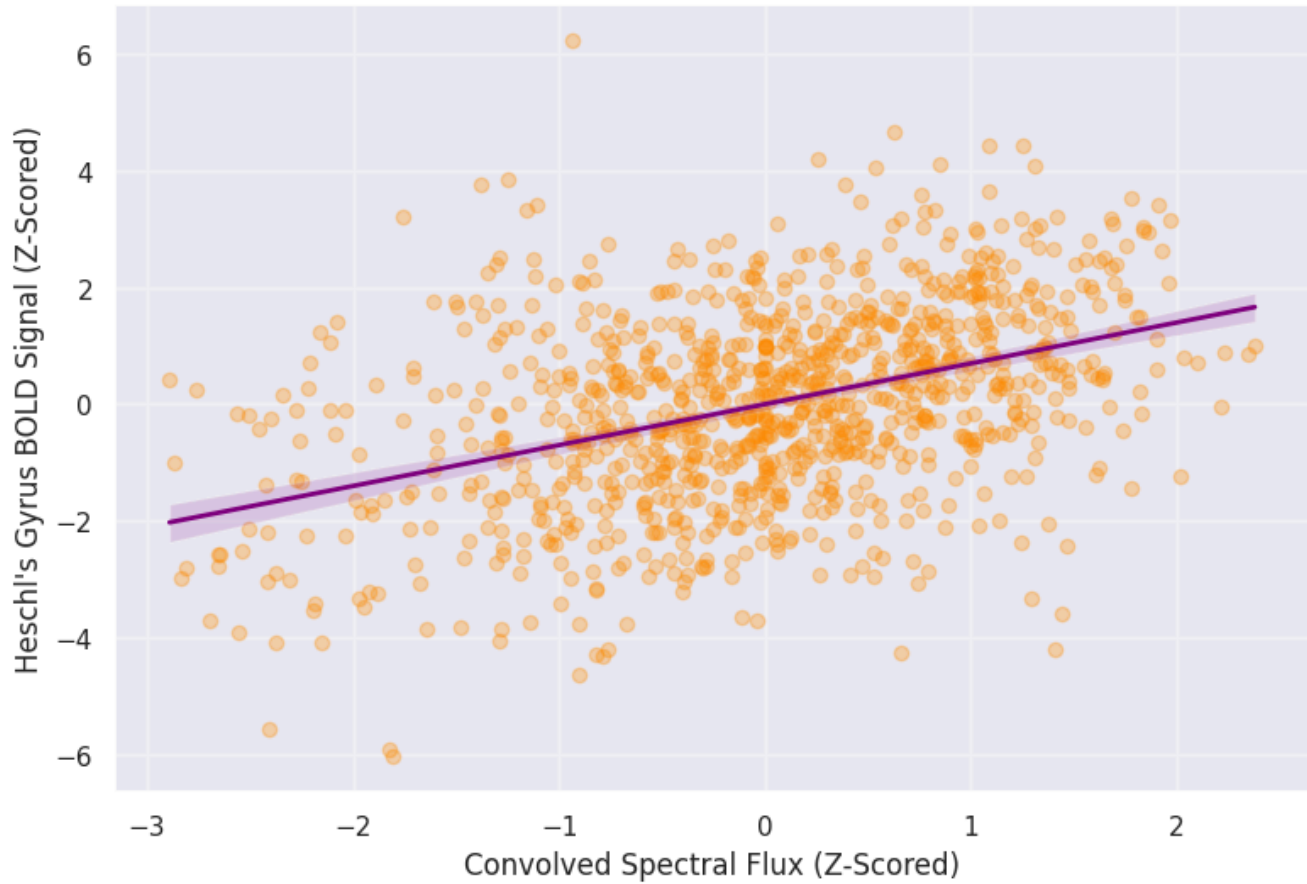
Now, let's do a similar analysis for spectral flux.



Overall A1 Correlation with Spectral Flux: $r=0.4125$, $p=5.5105e-44$

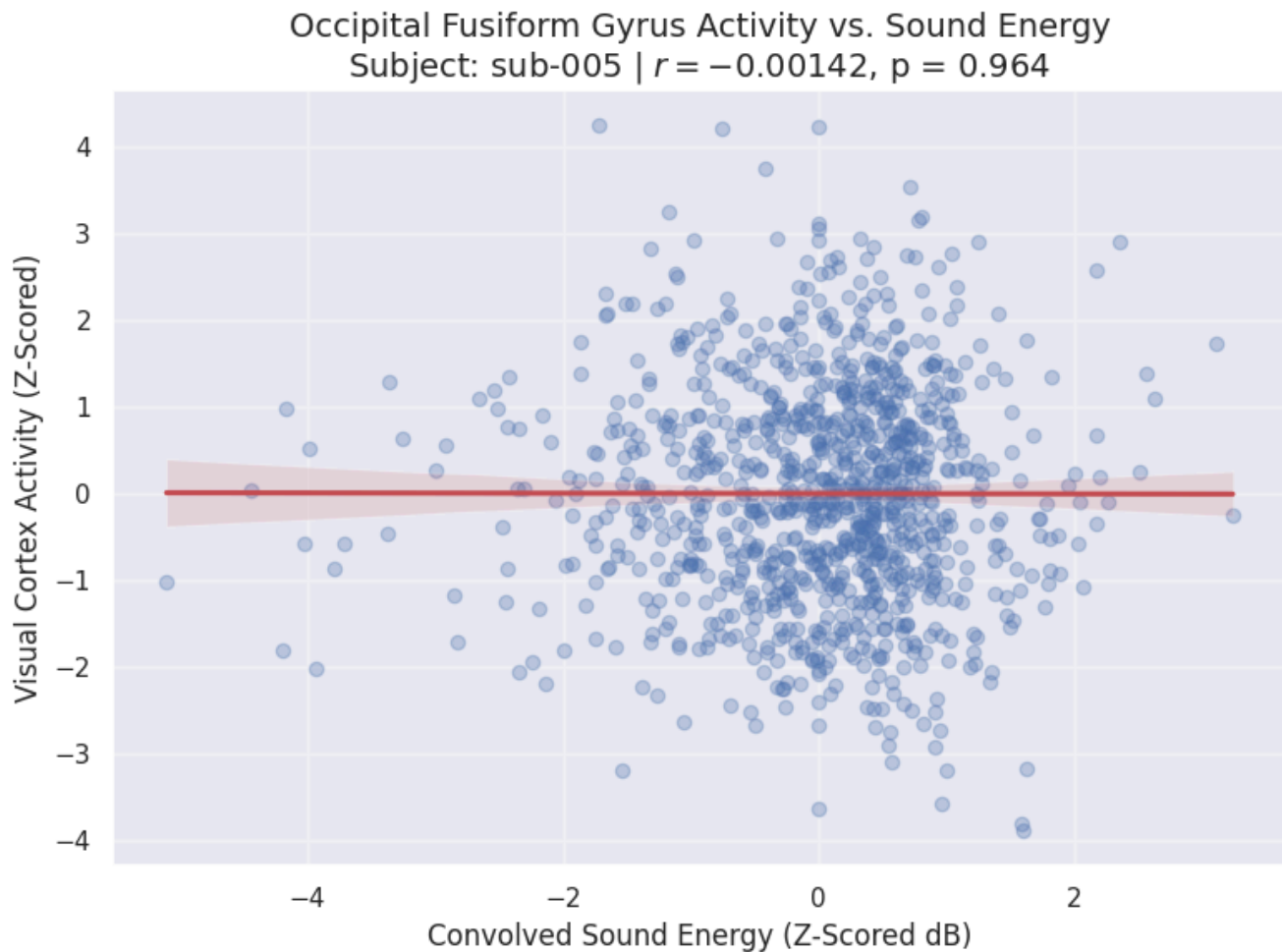
Heschl's Gyrus Activity vs. Spectral Flux

Subject: sub-005 | $r = 0.412$



One final check we'll do is to run the sound energy analysis on a control ROI. Here, we pick the Occipital Fusiform Gyrus, a key area of the visual cortex but not of the auditory system, in theory. We expect that it should have very little correlation with sound energy.

Corrected A1 Correlation: $r = -0.0014$, $p = 9.6361e-01$



Indeed, that's what we find! An almost horizontal line with r-value -0.00142 and p-value 0.964. With this initial confirmation of the validity of our audio parsing, we move on to novel analyses.

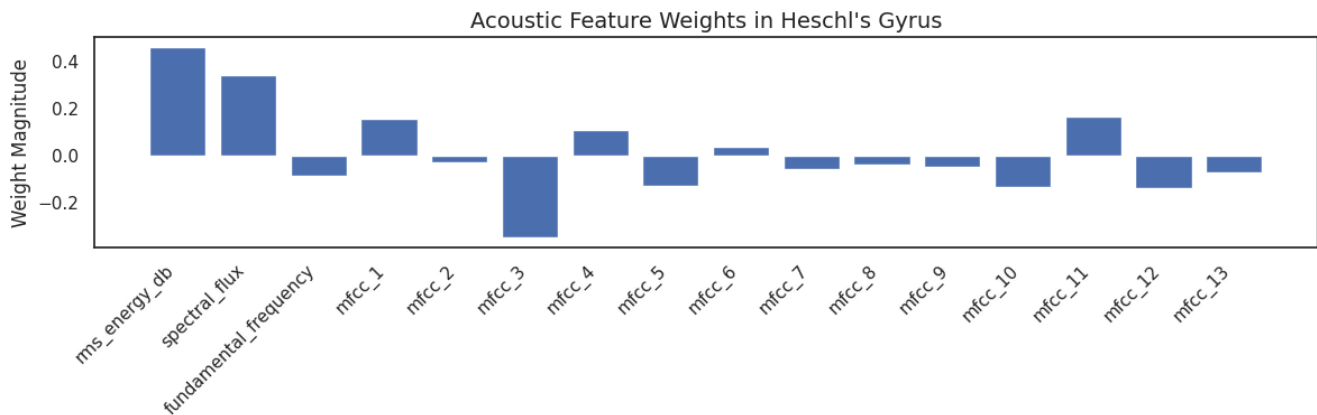
4. Classification

4A. Encoding Part 2

We want to see how well we can predict the brain's activity based on the continuous physical properties of the sound. Since we expect quantities such as sound energy and spectral flux to rise and fall together, we use L2 regularization as they are highly correlated. We first split into training and testing splits with a 90/10 split, standardize the training features, then cross-validate to find the best performing penalty parameter before testing it on the held-out data.

Encoding Model R^2 Score: 0.3683
Best Alpha: 46.42

The model score is not bad, but to see which features are driving the primary auditory cortex activity the most, we plot them below and see which weights (betas) are largest.



Unsurprisingly, sound energy and spectral flux seem to be the largest impact features, though it seems mfcc_3 also has a strong negative weight. MFCC3 generally captures mid-frequency peaks and valleys, typically associated with formant structures of human speech such as differentiating vowels. Since Heschl's gyrus does capture a lot of the finer details of human speech, this relationship makes sense.

In thinking about components such as fundamental frequency, we probably would not expect there to be a strong linear relationship between it and BOLD activity. That would imply that higher notes inspire higher activity overall, which intuitively does not make sense even as there may be brain regions that selectively respond to certain pitches.

SCRIPT FOR SAVING FILES

```
Saved final feature matrix.  
Saved raw inaSpeechSegmenter tags.  
Saved cleaned Gentle speech tags.
```

MISC/FOR LATER:

References

Atencio, C. A., Sharpee, T. O., & Schreiner, C. E. (2012). Receptive field dimensionality increases from the auditory midbrain to cortex. *Journal of neurophysiology*, 107(10), 2594–2603. <https://doi.org/10.1152/jn.01025.2011>.

Bendor, D., & Wang, X. The neuronal representation of pitch in primate auditory cortex. *Nature* 436, 1161–1165 (2005). <https://doi.org/10.1038/nature03867>.

Handwerker, D. A., Ollinger, J. M., & D'Esposito, M. (2004). Variation of BOLD hemodynamic responses across subjects and brain regions and their effects on statistical analyses. *NeuroImage*, 21(4), 1639-1651.

Hudspeth, A. J., & Konishi, M. (2000). Auditory neuroscience: development, transduction, and integration. *Proceedings of the National Academy of Sciences of the United States of America*, 97(22), 11690-11691. <https://doi.org/10.1073/pnas.97.22.11690>.

Nastase, S. A., Liu, Y.-F., Hillman, H., Zadbood, A., Hasenfratz, L., Keshavarzian, N., Chen, J., Honey, C. J., Yeshurun, Y., Regev, M., Nguyen, M., Chang, C. H. C., Baldassano, C., Lositsky, O., Simony, E., Chow, M. A., Leong, Y. C., Brooks, P. P., Micciche, E., ... Hasson, U. (2021). The "Narratives" fMRI dataset for evaluating models of naturalistic language comprehension. *Scientific Data*, 8(1), 250. <https://doi.org/10.1038/s41597-021-01033-3>

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Zatorre, R. J., Belin, P., & Penhune, V. B. (2002). Structure and function of auditory cortex: music and speech. *Trends in cognitive sciences*, 6(1), 37-46. [https://doi.org/10.1016/s1364-6613\(00\)01816-7](https://doi.org/10.1016/s1364-6613(00)01816-7).